Synthetic Data Augmentation for Mispronunciation Detection in L2 English

Ella Hawkins
ella.hawkins@maine.edu
University of Southern Maine
Portland, Maine, USA

Abstract

Mispronunciation detection and diagnosis is an important part of computer-assisted language learning systems, where accurate feedback on pronunciation can help learners improve without the need for a human tutor. Most previous approaches to this problem rely on acoustic features extracted from speech audio, which can make them sensitive to noise and speaker variation. This paper explores a different approach: treating mispronunciation detection as a sequence labeling problem over IPA transcriptions, rather than an acoustic classification task. Because annotated L2 transcription data is not readily available in large amounts, we also explore whether synthetically augmenting IPA sequences with errors based on known Mandarin L1 phonological patterns can improve model performance. We train and evaluate three sequence models on various amounts of synthetically augmented data. Results show that a moderate amount of synthetic data (4,000 samples) significantly improves phone- and word-level F1 for the BiLSTM and GRU, and that synthetic data at all evaluated scales significantly improves Transformer performance. Across all three models, synthetic augmentation reduces false positive error hallucinations, though it also tends to increase false negatives. These results suggest that IPA-based sequence modeling is a viable approach to mispronunciation detection, and that statistically-informed synthetic error generation can improve performance when real annotated data is scarce.

CCS Concepts

• **Computing Methodologies → Natural Language Processing**.

Keywords

Mispronunciation Detection, Data Augmentation

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1 Introduction

Mispronunciation detection and diagnosis (MDD) is an important component of computer-assisted language learning (CALL) systems, where it is used to provide learners with accurate feedback on their pronunciation. Most existing MDD systems approach this as an acoustic classification task, comparing a learner's speech against native pronunciation models using features extracted from the audio signal. While these approaches can be effective, they are sensitive to noise and speaker variation, and they can struggle to distinguish sounds that are phonologically distinct but acoustically similar (i.e., they are different sounds linguistically, but they can sound similar).

An alternative approach is to treat mispronunciation detection as a problem in phonological sequences rather than acoustic signals. The International Phonetic Alphabet (IPA) represents speech sounds symbolically, with each symbol corresponding to one phone (sound). If a learner's speech can be transcribed into IPA, sequence models can operate directly on these representations without any access to the underlying audio. This allows phonological knowledge to be incorporated more directly, and avoids some of the issues that come with acoustic methods. A key challenge with this approach is data scarcity. Annotated phoneme-level L2 transcriptions are difficult and expensive to obtain. To address this, we generate synthetic IPA transcriptions with errors injected based on statistics from a real Mandarin L1 learner dataset, and investigate whether training on this augmented data improves mispronunciation detection on real learner transcriptions.

This paper addresses the following research question:

> **RQ:** Does synthetic data augmentation improve mispronunciation detection performance on speech transcriptions of Mandarin L1 learners of English?

To answer this question, we train three sequence models (a BiLSTM, a GRU, and a Transformer) on IPA transcriptions from the speechocean762 dataset, with and without synthetic augmentation. Each model is evaluated at six training levels using phone-, word-, and sentence-level F1, and mean absolute error. By comparing models trained on real data only against models trained on both real and synthetic data at different scales, we can isolate the effect of synthetic data augmentation.

The rest of the paper is organized as follows. Section 2 reviews prior work on mispronunciation detection, synthetic data augmentation, and the use of phonological knowledge in MDD systems. Section 3 describes our datasets, synthetic error generation procedure, models, evaluation metrics, and baselines. Section 4 presents experimental results and discussion. In section 5 we discuss limitations and future work, and in Section 6 we conclude the paper.

2 Related Work

Previous research on mispronunciation detection and diagnosis (MDD) has explored the following areas relevant to this work: automated pronunciation error detection, synthetic data augmentation for MDD and automatic speech recognition (ASR) models, and the use of phonological knowledge to model English L2 speech.

Automated pronunciation error detection. Approaches to automated MDD can be categorized by the type of input they rely on: acoustic alone, acoustic-phonetic, linguistic-based and hybrid [13]. Acoustic-only approaches are based solely on features extracted from the speech signal. Acoustic-phonetic methods augment these features with phoneme-level information, often using forced alignment to capture deviations from native pronunciations [9, 14] . Linguistic-based approaches incorporate knowledge from phonology, such as IPA transcriptions or segmental rules, either as input features or as training constraints [6, 11]. Finally, hybrid approaches combine acoustic, linguistic, and phonetic knowledge, often with neural networks or Transformer-based models, to directly model pronunciation errors and improve detection accuracy [4, 15, 17]. Most MDD systems rely heavily on acoustic representations, which makes them sensitive to noise and speaker variation, and limits their ability to distinguish phonologically distinct but acoustically similar sounds [2, 17].

Synthetic Data Augmentation. Data scarcity is a challenge in MDD, since collecting and annotating real learner speech is costly. Previous work addresses this by using synthetic or augmented data. It generally uses one of two approaches, the first of which is to modify speech audio directly. This approach generates synthetic speech using text-to-speech or accent conversion systems, often guided by phonological knowledge, and uses the resulting waveforms to augment ASR or MDD training data [1–3, 7]. These methods have shown that synthetically accented speech can improve performance on seen accents. The second approach is converting speech to phonemes, modifying the phonemes rather than the audio, then converting those to graphemes (written representations). This approach values phoneme-level modeling for understanding pronunciation variation but typically revisits acoustics later on, leaving open the question of whether symbolic phonological representations alone can support mispronunciation detection [11, 19].

Phonological Error Modeling. Similar work examines how phonological knowledge can be used to mimic L2 pronunciation patterns, and how this data can be used to guide models. Most high-performing acoustic models learn phonological patterns (e.g., substitutions, deletions) implicitly from training data [9, 17]. This approach is shown to be effective, however, it requires large amounts of annotated data. Other approaches directly encode phonological insights and formalize those insights as rules to aid in data augmentation or to inform error analysis [11–13, 16, 18]. This approach is more data-efficient than the first approach, however, integration of the methods outlined by this approach into MDD systems remains largely unexplored.

To the best of our knowledge, no prior work has treated mispronunciation detection as a sequence labeling problem using IPA transcriptions, without access to acoustic data. Existing systems either operate only on audio, or use phonological representations as well as audio data. Similarly, prior work on synthetic data augmentation for MDD modifies audio, instead of speech transcriptions.

3 Methods

3.1 Real Data

The speechocean762 [20] dataset is a collection of annotated L2 English transcriptions. It is designed for pronunciation assessment use. It contains 5,000 sentence transcriptions from 250 different speakers, Mandarin being the first language for all of them. Each transcription is annotated at the sentence, word, and phone level. Sentences are scored 0-10 for accuracy, fluency, completeness and prosody. Words are scored 0-10 for accuracy and stress of the word's pronunciation. At the phone level, each phone is scored 0-2 for pronunciation accuracy. Where applicable, words are given mispronunciation metadata (see the structure of the sample below).

The word-level mispronunciation metadata, which is assigned to words that were pronounced with errors, records the canonical phone (the expected pronunciation), the pronounced phone (what the learner produced), and the phone's index in the word. A phone labeled with a pronounced value of "<DEL>" indicates that the phone was deleted. Annotations are provided in the ARPAbet, which is a phonetic transcription system similar to the IPA; to represent phones, it uses sequences of English letters instead of IPA symbols. We converted all ARPAbet transcriptions to the IPA using the Python phonecodes library.

Here is a sample simplified word JSON object:

```
{
  "text": "CLOUD",
  "phones": [k, l, aʊ, d],
  "mispronunciations": [
    {
      "canonical": aʊ,
      "index": 2,
      "pronounced": oʊ
    },
    {
      "canonical": d,
      "index": 3,
      "pronounced": "<DEL>"
    }
  ]
}
```

In this example, the word "cloud" (IPA: k l aʊ d) was pronounced with two errors: aʊ (the "ou" sound in cloud) was substituted with oʊ (the "oh" sound), and the d was deleted.

We split the speechocean762 dataset into training (80%), validation (10%), and test (10%) sets, stratified by sentence-level error count.

Figure 1 shows the score frequencies at both the word and sentence levels. The majority of words are scored as correct (score = 10), and most sentences are scored 8 or above, which shows that the majority of sentences and words were pronounced correctly. Figure 2 shows the 10 most frequently mispronounced phones; it is worth noting that half of these, including the top 2, are vowels. Figure 3 shows the frequencies of the positions of errors in words, with the position rounded to the nearest tenth. Errors are most

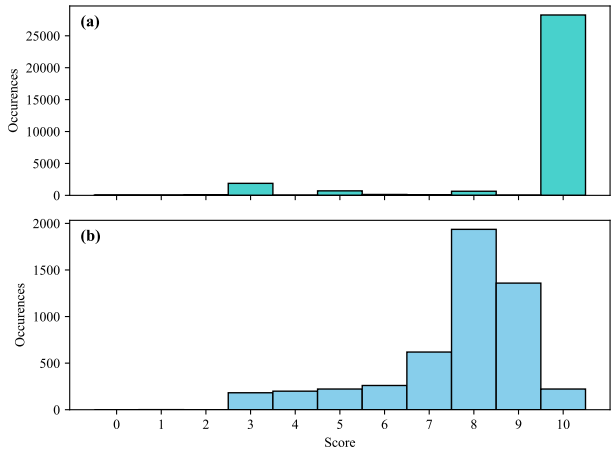

Figure 1: (a) Frequency of word-level pronunciation scores. (b) Frequency of sentence-level pronunciation scores.

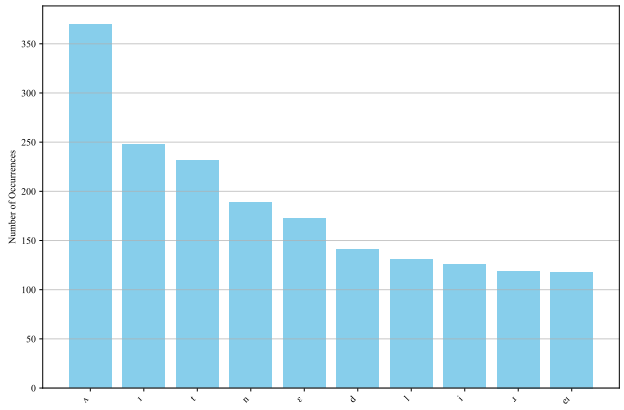

Figure 2: Top 10 frequently mispronounced phones.

often near the end of the word, specifically, they occur on the last phone. A fair number of errors also occur near the middle of the word. These statistics are referenced in the probability distribution used to inject the synthetic errors.

3.2 Synthetic Data

To generate synthetically augmented IPA transcriptions, we use the Jotschi/visual_genome-simple-en dataset on HuggingFace. This dataset contains simple English sentences produced by asking Mistral-7B-Instruct-v0.2-AWQ [5] to rephrase image captions from the Visual Genome dataset [8], which is a dataset of images from the Microsoft Common Objects in Context dataset [10], into a single sentence for a young child. Using LLM-generated sentences for children as a base is appropriate for this task because they are simple; the vocabulary and sentence structures are those commonly used in everyday spoken English, controlling the word- and structure-level complexities, which allows the focus to be on pronunciation patterns.

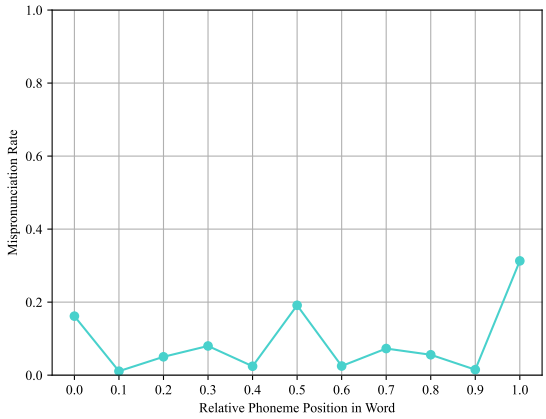

Figure 3: Mispronunciation rate as a function of phone position within word.

We convert these simple English sentences to the IPA as follows. Each sentence is first normalized to be fully capitalized, with punctuation removed. Then, we convert each word to the ARPAbet using the Carnegie Mellon Pronouncing Dictionary (CMUdict). Words not found in the CMUdict are handled by Python's g2p_en library. We do not use the G2P system for words already in the CMUdict to avoid inconsistencies. The ARPAbet transcriptions are then converted to the IPA using the phonecodes Python library, the same conversion method used for the real learner data, ensuring consistency across both datasets.

To inject errors into these transcriptions, we first compute statistics from the train set of the speechocean762 data. These statistics include the proportion of sentences with pronunciation errors, the average number of erroneous words in these sentences, phone error frequencies (see Figure 2), and the position of errors within words (see Figure 3). We then randomly select five sets of 2,000 sentences from the visual_genome-simple-en IPA transcriptions. Each sentence is probabilistically selected to contain errors based on the speechocean762 statistics. For sentences selected, one or more words are randomly chosen, and one or more phones in each word are sampled to contain an error, again based on the speechocean762 statistics.

3.3 Models

We use three sequence models (models that process ordered data): a gated recurrent unit (GRU), a bidirectional long short-term memory (BiLSTM), and a transformer. We use a shared vocabulary (of IPA tokens) across all models. Each model has two binary classification heads: a phone head, which predicts whether a phone in the sentence is an error, and a gap head, which, for each gap between IPA tokens, predicts whether a phone was deleted (i.e., if there should be a phone there, but the speaker did not produce it).

A GRU and a BiLSTM both have recurrent architectures that use context from both directions of the sequence (forward and backward). They both use gates to choose what to remember, but a GRU uses a simpler structure, while the BiLSTM uses a more

Table 1: Evaluation results for GRU. All models trained on 4,000 real samples. * indicates that the model did significantly better than the baseline.

Synthetic Samples	Phone F1	Word F1	Sentence F1	MAE
0 (baseline: 4,000 real)	0.3917	0.3827	**0.5185**	0.5920
2,000 synthetic	0.4122	0.3136	0.4516	0.5580
4,000 synthetic	**0.4601***	**0.3988**	0.5029	**0.5380**
6,000 synthetic	0.2072	0.2222	0.3583	1.7920
8,000 synthetic	0.2079	0.2188	0.3594	1.8660
10,000 synthetic	0.2095	0.2247	0.3683	1.7700

Table 2: Evaluation results for BiLSTM. All models trained on 4,000 real samples. * indicates that the model did significantly better than the baseline.

Synthetic Samples	Phone F1	Word F1	Sentence F1	MAE
0 (baseline: 4,000 real)	0.2806	0.2945	0.4541	1.0640
2,000 synthetic	0.2833	0.2864	0.4241	1.0720
4,000 synthetic	**0.4323***	**0.3940***	**0.4744**	**0.5640***
6,000 synthetic	0.1997	0.2158	0.3514	1.7080
8,000 synthetic	0.1665	0.1837	0.3345	2.2960
10,000 synthetic	0.1512	0.1625	0.3316	2.7700

complex structure, with separate memory. In contrast, a transformer uses self-attention instead of recurrence to consider the whole sequence at once.

All three models use a learning rate of 3×10^{-4}, a dropout of 0.1, a batch size of 32, and train on 20 epochs, selecting the best checkpoint based on phone-level metrics on the validation set. The GRU and BiLSTM both use an embedding dimension of 128, a hidden size of 256, and two layers. The transformer has a model dimension of 256, with 8 attention heads and 4 layers.

3.4 Evaluation Metrics

We evaluate each model at the phone, word, and sentence levels. The phone-level F1 is computed from binary error classification of each phone (correct = 0, error = 1). The word- and sentence-level F1 scores are computed similarly, treating any word with at least one phone error as an erroneous word, and each sentence with at least one phone error as an erroneous sentence. Additionally, we report the mean absolute error (MAE) between the predicted and ground-truth count of errors in each sentence.

Statistical significance is assessed for each training level against the real-only baseline using approximate randomization ($n = 10,000$). Training levels that performed significantly better (with $p < 0.05$) than the baseline on a specific metric are indicated by an asterisk.

3.5 Baseline Systems

To study the effect of synthetically augmented data on three different model types, we create a baseline for each model type. The baselines use the same architecture as the models trained with synthetic data, but they are trained only on the real learner data (speechocean762). Using the same architecture and evaluation metrics ensures a fair comparison, as all other experimental conditions remain identical, allowing us to isolate the effect of synthetic data augmentation.

The code for this project is available at: https://github.com/ella84211/synthetic-mispronunciation-data.

4 Experimental Results

We evaluate each model at six training levels: the real-only baseline (4,000 samples from the speechocean762 [20] dataset), and five conditions that use 2,000, 4,000, 6,000, 8,000, and 10,000 samples of synthetic data, in addition to the real data.

GRU Results. As shown in Table 1, the GRU trained on 4,000 synthetic samples performed better than the baseline GRU in all categories besides sentence-level F1. This is because the baseline tended to predict errors in sentences that actually had errors, but it often predicted them in the wrong spot. This caused the sentence-level correct prediction count to be higher than the phone- and word-level correct prediction counts. For instance, in the sentence, "he was face to face with the girl," which had errors in the first and last words, the baseline predicted only one error in the sentence, in the sixth word. This means that it correctly predicted the sentence as erroneous, but did not classify the phones and words correctly. Therefore, the sentence-level F1 score was higher than the phone- and word-level F1 scores.

The most frequently occurring pattern was that the GRU trained only on real data usually over-predicts errors, compared to the GRUs trained on synthetic data. For example, in the sentence "how long was he going to be here," the GRU trained on 4,000 samples of synthetic data correctly predicted one error in the first word and two errors in the second, however, the GRU trained on real data predicted three extra errors between two extra words, neither of which actually had errors. This same pattern is observed in the sentence "there are going to be more." While the GRU model trained on the synthetic data correctly predicts the two errors in the sentence, both of which appear in the first word, the GRU model trained only on the real data predicts three errors in the first and third words, and one error in the second word.

BiLSTM Results. Table 2 shows that training the BiLSTM on 4,000 samples of synthetic data significantly improves the MAE and the phone- and word- level F1 scores, compared to the real-only baseline. Training on 2,000 samples of synthetic data appears to have little effect on the results, and performance decreases when training on 6,000 synthetic samples or more.

Many of the instances where this BiLSTM performed better than the baseline occurred when the synthetic data BiLSTM correctly predicted a small amount of errors, but the baseline hallucinated a large number of errors. This pattern is also observed with the GRU models. For example, in the sentence, these were all little girls," the synthetic data BiLSTM predicted that there was one error in the first word. This prediction was correct. However, the BiLSTM trained only on real data predicted that there were two errors in the first word, and one error in each of the next two words, hallucinating a total of three errors. Another instance of this pattern is in the sentence, how long was he going to be here," the synthetic data BiLSTM correctly predicted one error in the first word and two errors in the second. However, the baseline predicted two errors in the first word, one error in the second, two errors in the fifth word, and two errors in the last word.

Table 3: Evaluation results for transformer. All models trained on 4,000 real samples. * indicates that the model did significantly better than the baseline.

Synthetic Samples	Phone F1	Word F1	Sentence F1	MAE
0 (baseline: 4,000 real)	0.0695	0.1192	0.3221	4.7140
2,000 synthetic	0.1909*	0.2016*	0.3834*	1.7980*
4,000 synthetic	0.2298*	0.2976*	0.4015*	0.9680*
6,000 synthetic	0.1093*	0.1402*	0.3221	3.5520*
8,000 synthetic	0.2804*	0.3435*	**0.4635***	**0.7760***
10,000 synthetic	**0.3048***	**0.3437***	0.4434*	0.7840*

On the other hand, the BiLSTM trained on 4,000 synthetic samples tends to under-predict errors. For example, in the sentence, Linda got the sunglasses," there was an error in the second word. The synthetic data BiLSTM did not predict any errors, however, the baseline correctly predicted this error. A similar pattern appeared in the sentence, there it was right next to him." The first word had three errors, however, the synthetic data BiLSTM predicted the sentence as correct. The real data BiLSTM correctly predicted these three errors. While the BiLSTM trained on 4,000 synthetic samples performs better than the real-only baseline, the BiLSTMs trained on 6,000, 8,000, and 10,000 synthetic samples tend to over-predict errors. For instance, in the sentence, "she wants to ask you herself," which did not have any errors, the BiLSTMs trained on more than 4,000 synthetic samples each predicted 3-5 errors, whereas the one trained on 4,000 synthetic samples correctly predicted none.

Transformer Results. The baseline transformer, trained on only real data, performs very poorly. It predicts errors in every sample, sometimes predicting whole words and sentences as erroneous. This is incorrect, as only about 20% of the sentences contain errors, of the 500 sentences in the test set. Additionally, of the sentences that contain errors, it is rare that full words and sentences are erroneous.

Table 3 shows that training the transformer on synthetic data at all evaluated scales significantly improves the performance compared to the baseline. Within the observed range (up to 10,000 samples), as the amount of synthetic data increases, the frequency of hallucinated errors decreases. For example, the sentence "I will put pressure on him and make him miss," was pronounced completely correctly. While the transformer trained on 10,000 synthetic samples correctly predicted that the sentence had no errors, the baseline predicted errors in every word besides the first, for a total of twenty-one errors, when none were pronounced.

Another example of this over-prediction is in the sentence, "it will have a body again," which had one error in the fifth word. The transformer trained on 10,000 synthetic samples correctly predicted that the word had an error, but it predicted that this error was at index 3, instead of index 1. However, this model performed better than the baseline, which predicted an error in the third word, and two errors in the fifth word. This shows that even when the baseline does not classify the majority (or all) of a sentence as erroneous, in most cases, it still predicts too many errors, whereas the model trained on synthetic data rarely has this issue.

Global Analysis. Two patterns were observed across all three models. First, training on the synthetic data helps all models prevent error hallucination; the models trained on both synthetic and real data consistently hallucinated fewer errors than the models trained on only real data. For the BiLSTM and the GRU, this is likely because the synthetic data allowed them to see more instances of errors in the synthetic data than what appeared in the real data. For the transformer, the synthetic data greatly improved the performance. As described above, there were multiple instances where the baseline transformer predicted every phone in a word as an error, for multiple words in a sentence, when there were no errors in the sentence at all. The transformer trained on synthetic data was able to avoid this problem. This is likely because of the transformer architecture; transformers often perform better when trained on larger amounts of data.

Table 4: Example model predictions for one word. 1 = error. B = baseline (real data only) and T = best synthetic condition (4,000 samples for GRU and BiLSTM; 10,000 samples for the transformer).

Phones	k	l	aʊ	d
Ground Truth	0	0	0	0
GRU B	**1**	**1**	0	0
GRU T	0	0	0	0
BiLSTM B	0	**1**	0	**1**
BiLSTM T	0	0	0	0
Transformer B	**1**	**1**	**1**	**1**
Transformer T	0	0	0	0

Second, while the models trained on synthetic data seem to hallucinate less, they tend to become more conservative when predicting errors. These models frequently under-predict errors, often failing to identify any errors in a sentence that contains several. A possible cause for this is that the synthetic data reduces the models' abilities to detect rare error patterns. The synthetic data BiLSTMs and GRUs often under-predict so much that they incorrectly classify whole words, predicting them as correct, even when every phone in the word is pronounced with an error. This pattern continues with the transformers, with the baseline transformer predicting too many errors, often at incorrect positions, and many of the synthetic data transformers not predicting any errors. A possible cause for this is that the synthetic data reduces the models' abilities to detect rare error patterns, as the errors in the synthetic data are injected based on probability. This means that the majority of the errors in the synthetic data are more frequent errors; therefore, the model may not see a large range of uncommon patterns.

Table 4 shows example predictions for each model type (the real data baseline and the best-performing model trained on synthetic data) on the word "cloud" (IPA: k l aʊ d), which was pronounced correctly (with no errors). "0" indicates that the model predicted the phone to be correct, and "1" indicates that the model predicted the phone to be an error. The transformer baseline does the worst; it predicts every phone to be an error. The GRU and BiLSTM baselines also hallucinate errors that do not exist. However, all three of the models trained on synthetic data correctly predict that the word was pronounced correctly.

Table 5: Sentence-level false positive (FP) and false negative (FN) error counts by model. B = baseline (real data only) and T = best synthetic condition (4,000 samples for GRU and BiLSTM; 10,000 samples for the transformer).

Model	FP	FN	Total Errors
GRU B	190	12	202
GRU T	35	52	87
BiLSTM B	90	12	102
BiLSTM T	68	45	113
Transformer B	404	0	404
Transformer T	69	49	118

Error Analysis. Errors in the models' predictions are classified as either false positive or false negative. A false positive error means that the model predicted an error where there were none, and a false negative error means that the model did not predict an error where one existed. For each model type, we consider the baseline (trained only on real data) and the best-performing model of that type (trained on synthetic data).

As shown in Table 5, for all models, with the exception of the GRU trained on 4,000 samples of synthetic data, the majority of the models' prediction errors were false positives. The baseline transformer model predicts the most errors, of the models evaluated. Its prediction was incorrect for 404 sentences. All of these predictions were false positives. This shows that the baseline transformer almost always predicts that errors exist, even when there are none.

Similarly, the BiLSTM and GRU baselines also show a tendency towards false positives, but they predict errors less aggressively. They each predicted a number of sentences to have no errors, unlike the transformer, which predicted that most sentences had errors. The predictions from both the BiLSTM and the GRU had a fair number of both false positive and false negative errors, but they both still had far more false positives than false negatives.

Training with synthetic data greatly changes this error distribution. For the BiLSTM trained on 4,000 samples of synthetic data, the false positive errors were reduced from 90 (from the BiLSTM trained only on real data) to 68. However, the false negative errors increased from 12 to 45. The GRU follows a similar pattern; the false positives were reduced from 190 to 35, but the false negatives were increased from 12 to 52.

5 Discussion

The high number of false positives from the baseline models' predictions indicates that training on limited real data makes it difficult for models to detect when a sentence is pronounced correctly. This suggests that additional data is necessary for the models to better learn pronunciation error patterns. While synthetic data does improve this issue, the increase in false negatives across all three models trained on synthetic data suggests that the synthetic data may not realistically represent more uncommon error patterns. Overall, these results suggest that synthetic augmentation does improve mispronunciation detection performance, though the effect varies by model and scale.

One limitation of our work is that the synthetic data does not represent more uncommon errors or error context. The errors are injected into the synthetic data based on the frequencies that they appear in the speechocean762 data. There are certain errors (such as removing an "s") that occur more often before or after a specific phone. For example, the "s" is deleted more often after consonants than vowels. In this study, many models are not learning the context because it is not in the synthetic data; including the context could help them have more accurate predictions. Future work could include oversampling more uncommon errors and having a more complex system that considers context, or developing a rule-based approach using linguistic knowledge for synthetic error injection.

Another limitation is that the synthetic data does not imitate error clusters in words and sentences. Therefore, the models are learning that errors are somewhat evenly distributed, whereas in real learner speech, errors are likely to be clustered, with many errors appearing close to each other in both words and sentences. Future work could include studying the attributes of these clusters and developing a system to make the appearance of the synthetic errors more realistic.

6 Conclusion

This paper explored mispronunciation detection as a sequence labeling problem over IPA transcriptions, with synthetic data augmentation used to address the scarcity of annotated L2 learner data. We trained three sequence models (a GRU, a BiLSTM, and a Transformer) across six training conditions, varying the amount of synthetically augmented data generated from Mandarin L1 phonological error statistics from the speechocean762 dataset. The results support our hypothesis: synthetic augmentation improves mispronunciation detection performance over the real-data-only baselines. For the GRU and BiLSTM, 4,000 synthetic samples produced the best results, with statistically significant improvements in phone- and word-level F1. For the Transformer, improvements were observed at all evaluated scales and generally increased as more synthetic data was added. Across all three models, the most consistent effect of synthetic augmentation was a reduction in false positive error hallucinations, but this came at the cost of more false negatives; this is likely because the synthetic data over-represents common errors and under-represents rare ones.

These findings suggest that phonological sequence modeling is a viable approach to mispronunciation detection, and that statistically-informed synthetic error generation is a useful tool for addressing data scarcity in this area. Future work should focus on improving synthetic error generation to better capture error context, clustering, and rare patterns, and on exploring whether incorporating lightweight acoustic features alongside IPA representations could improve performance further.

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